

- (f) Indien $R = 19,6$ m, bepaal die vlugtyd van die projektiel.
 If $R = 19,6$ m, determine the time of flight of the projectile.

$$t_p = \sqrt{\frac{2R}{g}} = \sqrt{\frac{2(19,6)}{9,8}} = 2 \text{ s}$$

(3)

- (g) Toon aan dat die speed van die projektiel by die toppunt van die baan gegee word deur $v_{\text{top}} = \frac{1}{2}\sqrt{3gR}$.

Show that the speed of the projectile at the top of the trajectory is given by $v_{\text{top}} = \frac{1}{2}\sqrt{3gR}$.

$$v_{\text{top}} = \frac{1}{2}\sqrt{3gR}$$

(2)

- (h) Toon aan dat die hoek waarteen die projektiel gelanseer word is $33,7^\circ$.
 Show that the angle at which the projectile has been launched is $33,7^\circ$.

$$\sin \theta = \frac{v_{\text{top}}}{v_{\text{initial}}} = \frac{\frac{1}{2}\sqrt{3gR}}{\sqrt{2gR}} = \frac{1}{2}\sqrt{\frac{3}{2}} = 0,612 \Rightarrow \theta = 33,7^\circ$$

(4)

[25]

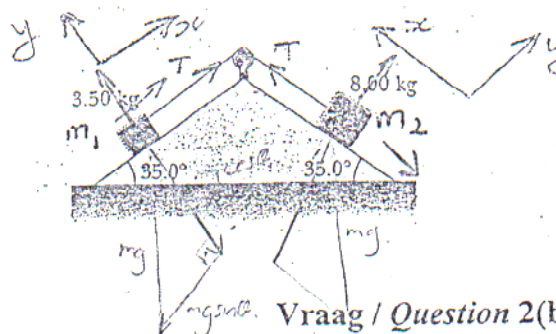
VRAAG / QUESTION 2

- (a) Met μ_k die kinetiese wrywingskoëffisiënt, skryf 'n uitdrukking neer vir die kinetiese wrywingskrag f_k .
 The coefficient for kinetic friction is μ_k . Give an expression for the force of kinetic friction f_k .

$$f_k = \mu_k N$$

(1)

- (b) Twee blokke met massas van $3,5$ kg en $8,0$ kg lê op wrywinglose vlakke en is verbind soos in die skets aangetoon word. Gebruik die assestelsels vir elk van die twee vlakke en:
 Two blocks of mass $3,5$ kg and $8,0$ kg are connected by a string on frictionless inclines. Use the axis on each incline as shown in the figure and:



Vraag / Question 2(b)(i)(Vervolg)(Continue) /